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Second-Year Internship Report:

INRIA Paris-Saclay: BOOST Team



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I - Introduction

a) INRIA

Created in 1967, Inria (French National Institute for Research in Digital Science and Technology) is under the joint authority of the Ministry of Higher Education and Research and the Ministry of the Economy. It has around 3,900 employees, including 2,800 scientists, spread across 9 research centres in France. The institute is organized around 220 project teams, often in partnership with universities and grandes écoles (École Polytechnique, CentraleSupélec, Télécom Paris, etc.), fostering interdisciplinary collaborations.

Inria's mission is to produce cutting-edge scientific results, contribute to France's digital sovereignty, and ensure technology transfer to industry. Each year, the institute supports more than 150 start-ups and innovative projects and publishes around 3,000 scientific papers.

Some emblematic projects:

- Introduction of the Internet in France and creation of the Renater academic network.
- Scilab: open-source scientific computing software.
- Coq: mathematical proof assistant for critical software.
- OCaml: functional programming language used by companies such as Facebook and Jane Street, and taught in particular in French preparatory classes.
- CompCert: certified C compiler for critical systems.
- Start-ups originating from Inria: Dataiku, Alan, Scalify.

I discovered this institute through the second-year engineering project, during which an Inria Nancy start-up was our client for the development of a camera system designed to improve patient flow in emergency departments. Having greatly enjoyed this experience, I wanted to repeat it by completing my internship within this institute. It was Inria Saclay, and more specifically the BOOST team, that kindly agreed to host me for these two months.

b) The Inria Centre at Université Paris-Saclay

The Inria Centre at Université Paris-Saclay is one of the institute's strategic hubs. Created in 1983 under the name Inria Rocquencourt, it was renamed the "Inria Centre at Université Paris-Saclay" in 2015 to reflect its integration into the new scientific ecosystem of the Saclay plateau. Since 2017, it has been fully established on the Université Paris-Saclay campus, alongside École Polytechnique, CentraleSupélec, ENS Paris-Saclay, Télécom Paris and numerous laboratories.

The centre brings together nearly 600 researchers, engineers and doctoral students, organized into around 40 project teams. Its research topics cover a wide spectrum: artificial intelligence, machine learning, cybersecurity, quantum computing, robotics, high-performance computing and embedded systems.

Thanks to its proximity to innovative companies (start-ups, SMEs and large groups), the centre plays a key role in technology transfer and participates in numerous collaborative projects funded by the European Union and the French National Research Agency. It also hosts many interns and doctoral students each year, making it a privileged place for research-based training.

c) The BOOST Project Team

The BOOST project team (Bio-informed mOnitoring & Optimization for enhanced Sport & healTh) is a joint initiative of Inria and Université Paris-Saclay, launched in January 2025. It focuses on the analysis of physiological signals obtained from connected devices, such as photoplethysmography (PPG), to assess the health of professional athletes.

Scientific objectives:

BOOST develops algorithms capable of interpreting physiological data such as electrocardiograms (ECG), electroencephalograms (EEG), blood pressure and PPG. The objective is to extract relevant indicators for assessing health and stress management in athletes, by combining signal processing, mathematical modelling and artificial intelligence.

Team and organization:

The team is led by Taous-Meriem Laleg and brings together 7 permanent researchers, 8 doctoral students, as well as many interns. It works closely with the CIAMS laboratory (Complexité, Innovation et Activités Motrices et Sportives) at Université Paris-Saclay, combining expertise in neuroscience, physiology, modelling and sport sciences.

CIAMS and the premises:

CIAMS hosts laboratories dedicated to applied research on motor and sporting activities. During my internship, I had the opportunity to attend the inauguration of the BOOST team's new premises at CIAMS, which allowed me to discover all the projects being carried out and the organization of the teams within this collaborative space.

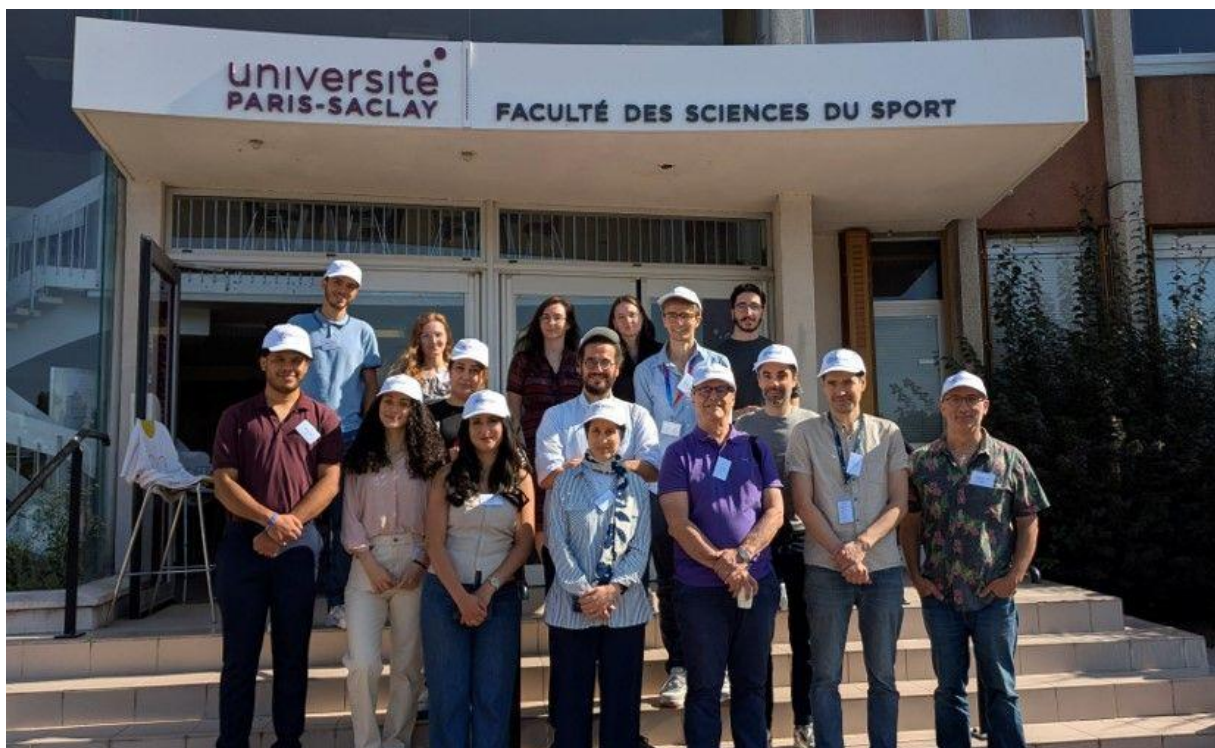


Figure 1: inauguration of the BOOST team's new premises at CIAMS

d) Work organization and working methods

Team members benefit from a high degree of flexibility regarding their working hours, with flexible arrival and departure times and sometimes one day of remote work during the week. A weekly meeting takes place every Tuesday morning, during which each member presents the progress of their project and receives questions and advice from the others. Having attended each of these meetings, I was able to observe the highly advanced level of the discussions, to the point of sometimes feeling out of my depth during the presentations.

The summer period, combined with this flexibility, meant that supervisors were sometimes present irregularly, which made project follow-up more difficult. I therefore felt a certain lack of structure, which could have made the integration and supervision of interns easier. It was not unusual to spend several days without seeing anyone in the offices.

However, as the team is still recent, Ms. Laleg had to manage many administrative appointments; I am convinced that this aspect will improve over time.

The team also has an educational role: during my internship, we welcomed high-school students on discovery internships to introduce them to the world of research and to our work.

II - Immersion in research work: PWV from the PPG signal

The first phase of my internship mainly consisted of exploring the tools and methods used by the team, as well as studying past and ongoing projects in depth, in order to acquire the level required to understand and, in a second stage, contribute effectively to one of them. The objective was therefore less to master one specific topic than to acquire a solid technical and scientific foundation.

During this first month, I analyzed a large amount of internal documentation and scientific papers, and I worked with several key tools (for example: EBM, visibility graphs, SCSA, etc.). Although I had previously taken artificial-intelligence courses as part of my curriculum, these were still far from the level required to work immediately on the team's projects, which involve cutting-edge AI. This preparatory stage therefore allowed me to bridge part of this gap and become familiar with approaches that were much more advanced than those covered in my initial training.

To put this learning into practice, I chose to dissect one project in particular: **"Assessment of Pulse Wave Velocity through Weighted Visibility Graph Metrics from Photoplethysmographic Signals"**, which I will detail below. This analysis served as a common thread for understanding the team's scientific approach and learning to use its tools.

Representing nearly half of my internship, this phase of becoming familiar with the tools and methods was a key period and deserves an important place in this report.

a) Context

High blood pressure affects around one adult in three worldwide, and better control of blood pressure could prevent more than ten million deaths each year. Among early indicators of cardiovascular risk, Pulse Wave Velocity (PWV) stands out as a reliable biomarker of arterial stiffness. PWV corresponds to the speed at which the pressure wave generated by contraction of the heart propagates along the arteries. The stiffer the arteries, the faster the blood travels, which is associated with cardiovascular risks such as hypertension, atherosclerosis or cardiac events.

The value of PWV lies in its ability to detect cardiovascular abnormalities at an early stage, before clinical symptoms appear. A high PWV can therefore warn of excessive arterial stiffness and make it possible to adapt treatment or lifestyle habits in order to prevent serious complications.

Conventional methods for measuring PWV are reliable but have important limitations. They require the simultaneous acquisition of the pulse signal at at least two distinct points on the body, a technically complex operation that is often operator-dependent. These constraints limit the frequency and accessibility of these measurements, particularly outside the hospital setting.

Faced with these limitations, the team became interested in non-invasive methods based on photoplethysmography (PPG). PPG consists of emitting a red or infrared light beam onto a highly vascularized area (such as the wrist or finger) and measuring the light reflected by the tissues. The amount of light absorbed varies with blood flow: it decreases when blood passes through and increases during the recession phase. By recording these variations several hundred times per second, it is possible to follow the cardiac cycle precisely. Many consumer devices, such as smartwatches, rely on this principle to measure heart rate and other vital signs. Estimating PWV

from PPG therefore offers a practical and promising tool that could be integrated into the connected-health devices of tomorrow.

The objective of my internship was to understand and master the complete pipeline for estimating PWV from PPG signals, so that I could later reuse this methodology in a more applied project.

b) Data used

Two types of data were used to estimate Pulse Wave Velocity (PWV) from PPG signals: real data obtained from human participants and simulated data generated by an in-silico model.

Real data:

The real signals came from 35 participants aged 35 to 73. Recordings were made on the radial artery at the wrist, with a sampling frequency of 200 Hz. Each acquisition lasted less than five minutes in order to minimize participant discomfort. The collected PPG signals naturally contain physiological noise, an essential element for testing the robustness of PWV-estimation methods under conditions close to clinical practice.

Each PPG signal is associated with a carotid-femoral PWV measurement obtained with the SphygmoCor device. These PWV values serve as the reference for evaluating the performance of the methods developed.

Simulated data (in-silico)

In addition, simulated signals were generated from an in-silico model representing 4,374 virtual adults aged 25 to 75, divided into ten-year intervals. The heart rates of the simulated subjects ranged from 66 to 86 beats per minute, and the signals were generated at a sampling frequency of 500 Hz.

These simulated signals are smoother and more idealized than the real signals and contain the theoretical PWV values as well as other cardiovascular parameters (arterial diameter, blood density, etc.). They provide a controlled environment for validating and training the model without the physiological noise inherent in human data.

c. Signal pre-processing and derivative calculation

Before extracting features for PWV estimation, the PPG signals required pre-processing in order to reduce noise and optimize the information contained in the signals.

1. Data filtering:

PPG signals are sensitive to different types of noise, particularly movement and respiratory variations. To improve signal quality, an 18th-order Chebyshev II band-pass filter was applied, with a low cut-off frequency of 0.5 Hz and a high cut-off frequency of 10 Hz. This step is very common for cardiac signals and makes it possible to preserve essential physiological information while eliminating irrelevant slow or fast fluctuations.

2. Derivative calculation

After filtering, the first and second derivatives of the PPG signals were calculated. These derivatives provide complementary information on the dynamics of the pulse wave:

- The first derivative highlights the rate of change of the signal, allowing the rise and fall of the pulse wave to be tracked more precisely.
- The second derivative highlights inflection points and subtle variations in waveform shape, capturing fine information about arterial stiffness.

The use of derivatives is a crucial aspect of the pipeline because they enrich the input signal for the model and improve PWV-estimation performance, as observed in the literature.

d. Transformation of signals into images: Visibility Graph

To make the best possible use of the information contained in the PPG signals and their derivatives, a signal-to-image transformation was performed using a Visibility Graph (VG). This method converts a time series into a graph, where each point in the signal becomes a node and connections between nodes are established according to a visibility criterion.

1. Principle of visibility

The two points of the signal are considered visible to one another if no intermediate value blocks the line of sight, that is, if for every point with the following condition is met:

$$y_c < y_a + ((y_b - y_a) / (t_b - t_a)) * (t_c - t_a)$$

The time series can be imagined as a hill, where each point corresponds to a location on that hill. Each point is therefore a "hiker". Two points are visible to one another if the hikers can look directly at each other without the intermediate terrain (the other points in the series) blocking their line of sight. If this is the case, a connection is created between these points in the visibility graph.

This approach makes it possible to capture temporal relationships between points in the signal, whether rapid variations within a cycle or longer-term relationships between successive cycles.

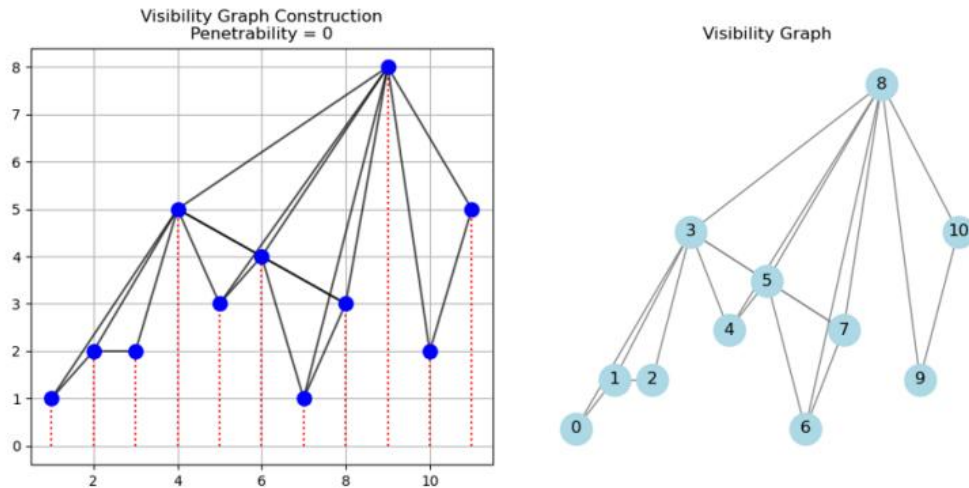


Figure 2: example of a visibility graph

2. Adjacency matrix and image

From the established connections, an adjacency matrix A of size $T \times T$ (where T is the number of points in the signal) is constructed:

$$A_{i,j} = \begin{cases} 1 & \text{if points } i,j \text{ are visible} \\ 0 & \text{otherwise} \end{cases}$$

This matrix can then be interpreted as an image, to which image-processing techniques can be applied in order to extract features.

The adjacency matrix A obtained from the visibility graph is symmetric because the visibility criterion is reciprocal: if a point P_i sees P_j , then P_j also sees P_i . Thus,

$$A_{i,j} = A_{j,i}, \forall i,j$$

This symmetry means that the matrix is diagonalizable, and a great deal of information can be extracted from its eigenvalues.

The diagonal of the matrix contains only 1s, corresponding to the visibility of a point with itself:

$$A_{i,i} = 1$$

- A column of 1s in the matrix may indicate a peak in the signal.
- 1s far from the diagonal reflect visibility between two distant peaks, representing long-term relationships in the signal.
- Conversely, a cluster of 1s close to the diagonal represents short-term relationships, corresponding to rapid events within a cardiac cycle.

3. Weighted graph

To enrich the representation of the signal, different weights can be applied to the edges of the visibility graph. These weightings make it possible to capture the morphology and dynamics of the signal more finely, beyond simple binary connectivity. The image produced from the adjacency matrix can therefore be coloured, reflecting the richness of the relationships between points.

If P_i and P_j are visible, the "1" in the adjacency matrix can be replaced by different types of weights, the most common being:

Euclidean distance:

$$EC_{i,j} = \sqrt{(y_j - y_i)^2 + (t_j - t_i)^2}$$

Squared Euclidean distance:

$$SEC_{i,j} = (y_j - y_i)^2 + (t_j - t_i)^2$$

Vertical distance and absolute vertical distance:

$$V_{i,j} = y_j - y_i, AV_{i,j} = |y_j - y_i|$$

Horizontal distance and absolute horizontal distance:

$$H_{i,j} = t_j - t_i, AH_{i,j} = |t_j - t_i|$$

It should be noted that this information is already visible in the unweighted adjacency matrix by looking at the proximity to the diagonal.

Slope and absolute slope:

$$S_{i,j} = \frac{y_j - y_i}{t_j - t_i}, AS_{i,j} = \left| \frac{y_j - y_i}{t_j - t_i} \right|$$

Each type of weighting emphasizes a particular aspect of the signal:

- Distances capture temporal and amplitude relationships,
- Slopes represent local variation in the signal, which is essential for analyzing rapid changes in cardiac cycles.

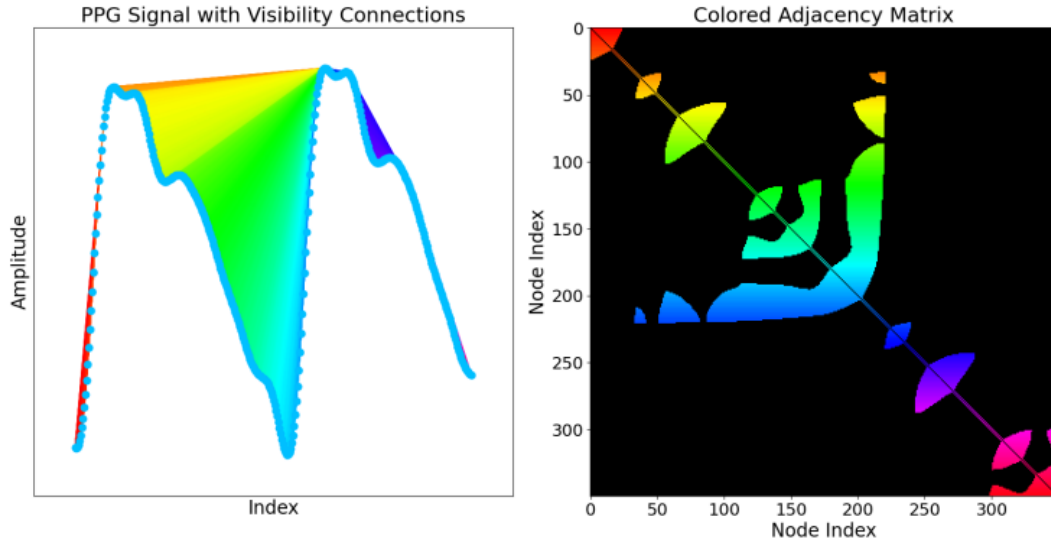


Figure 3: adjacency matrix of a visibility graph in which each edge has a different colour.

4. Penetrability parameter

A parameter p controls the penetrability of the graph, i.e. the number of intermediate points that may be ignored in order to establish a connection. This makes the graph more robust to noise by preserving meaningful relationships even in the presence of physiological fluctuations or noisy measurements.

Each signal (raw and derivatives) is therefore transformed into several images representing different perspectives on blood-flow dynamics. These images then serve as the basis for image-feature extraction, detailed in the following section.

e. Feature extraction from images

Once the PPG signals and their derivatives have been converted into images using the visibility graph, the next step is to transform these images into measurable numerical values. These values, called features, constitute the input to the PWV-estimation model. This step is essential: the model does not analyze the images directly, but rather the numbers that describe their shape, texture and structure.

We therefore built a pipeline based on four complementary families of features, each capturing a specific aspect of the image:

1. Shape features

Hu moments

- Hu moments condense the overall shape of an image into 7 numerical values that are invariant to rotation, translation and scaling.
- In practical terms, these values describe how the pixels are distributed in the image. For example, a highly symmetrical image will produce different moment values from a highly asymmetrical image.
- This allows the model to compare signals even if their amplitude or position has varied slightly.

Zernike moments

- Based on orthogonal polynomials, they provide a finer and more localized analysis than Hu moments.
- They decompose the image into circular patterns (somewhat like a wave decomposition) and provide coefficients that measure the presence of these patterns.
- This approach captures subtle details in cardiac cycles, such as small variations in signal amplitude or shape.

2. Morphological features

Texture Analysis Statistics (TAS)

- The image is divided into 3 x 3 pixel patterns.
- For each small pattern, statistics are calculated (uniformity, contrast, energy, entropy, etc.).
- This makes it possible to quantify how the local texture of the image varies: smooth areas, irregular areas, abrupt transitions, etc.
- This is very useful because the visibility graph produces repetitive patterns that directly reflect the regularity or irregularity of heartbeats.

3. Frequency features

Wavelet Packet Transform (WP): 2D wavelet-packet decomposition makes it possible to analyze texture at different resolutions and orientations. WP coefficients are used to calculate statistical measures (mean, standard deviation) reflecting the micro- and macro-textures of the image.

4. SCSA features (Semi-Classical Signal Analysis)

This stage of the pipeline is quite unique because it draws directly on quantum physics. Semi-Classical Signal Analysis (SCSA) uses principles close to the Schrödinger equation to analyze the signal. In practical terms, the matrix obtained from the visibility graph is interpreted as a "landscape" on which eigenvalues are calculated, somewhat as one would do for energy levels in a quantum system.

From these eigenvalues, spectral invariants (INV1, Kn, En, etc.) are extracted, which summarize the overall structure of the signal in a very robust way. These indicators are relatively insensitive to noise and complement the other classical image features well.

In summary, SCSA provides a global and stable view of the PPG signal and its derivatives, while introducing into the pipeline a tool that is rarely used in biomedicine and comes from the world of physics, making it an original but very complex asset. Although I now know how to use it, the reasons why it works still seem very obscure to me.

f. Feature selection

Each image obtained from the visibility graph contains several hundred features extracted both from the raw signals and from their derivatives. However, not all of these features are necessarily informative for PWV estimation. Some may be redundant or introduce noise, which can slow down model learning and reduce its ability to generalize to new signals.

To identify the most relevant features, we proceeded as follows:

1. Evaluation of correlation with PWV:

Each feature was compared with the corresponding PWV value.

Four different methods were used to capture different types of relationships:

- Pearson: to detect linear relationships between the feature and PWV.
- Spearman and Kendall: to detect monotonic relationships, whether linear or not.
- Xicor: to detect more complex and nonlinear relationships.

2. Automatic feature selection:

For each method, the top K most highly correlated features were retained.

To avoid duplicates, any feature appearing in several methods was retained only once.

3. Advantages of this selection:

Reduction in redundancy: only genuinely informative features are retained.

Improved learning speed: the model processes less unnecessary data.

Improved generalization: the model is less sensitive to noise and random fluctuations, making it possible to predict PWV more accurately on new signals.

This selection step therefore makes it possible to focus learning on the most relevant information while keeping the model simple and robust.

g. Machine learning model

To predict PWV from the selected features, we used an Explainable Boosting Machine (EBM) model. This model belongs to the family of Generalized Additive Models (GAMs), whose principle is to represent the output as the sum of simple functions applied to the features. The major advantage of this approach is that it combines predictive accuracy and interpretability, which is essential in a medical context where it is necessary to understand why the model makes a particular prediction.

How the EBM works:

Each feature contributes individually to the PWV prediction. The model estimates an influence function for each feature, making it possible to visualize how a high or low value of that feature affects the predicted PWV.

Some interactions between features can also be included, for example to capture complex combined effects that would not be visible by looking at a single variable.

The EBM uses shallow decision trees to estimate these functions, then combines these trees in a boosted manner, i.e. by progressively building trees that correct the errors of the previous ones.

Advantages:

The contributions of each feature can be visualized and interpreted directly, making it possible to identify which features of the PPG or its derivatives are the most informative for PWV.

The model is robust to overfitting thanks to boosting, while remaining simple enough to be explained and understood, unlike a complex neural network (MLP or CNN).

h. Model training

Training the EBM model requires several stages in order to ensure reliable and robust predictions.

1. Feature normalization

Before learning, all selected features are normalized so that they have a mean of zero and a variance equal to 1 (z-score).

This puts all features on the same scale, preventing variables with larger values from dominating learning.

2. Cross-validation (5-fold cross-validation)

To evaluate the model's ability to generalize to new data, 5-fold cross-validation is applied:

- The dataset is divided into 5 equivalent subsets.
- At each iteration, 4 subsets are used for training and 1 for evaluation.
- This process is repeated 5 times, with each subset serving exactly once as the test set.
- This method limits the risk of overfitting and provides a more reliable estimate of the model's actual performance on unseen data.

3. Hyperparameter optimization

To maximize EBM performance, the hyperparameters (for example the number of trees, tree depth and learning rate) are optimized using the TPE (Tree-structured Parzen Estimator) method from the Optuna library:

TPE is a Bayesian optimization method that intelligently explores the hyperparameter space based on previously tested performance.

The objective is to find the parameter combination that minimizes the model's prediction error while maintaining its robustness.

i. Model evaluation

To ensure that the EBM model predicts PWV correctly and does not learn coincidences or noise, two types of tests are performed: positive tests and negative (permutation) tests.

1. Positive tests

These tests consist of applying the model to the real test set, i.e. to PPG signals and derivatives that the model has never seen during training.

Performance is quantified using two metrics:

- R^2 (coefficient of determination): measures the proportion of PWV variance explained by the model. A value close to 1 indicates that the model captures the real variations in PWV well.
- RMSE (Root Mean Square Error): measures the average difference between predicted and actual values. A low RMSE indicates accurate predictions.

These metrics make it possible to verify that the model provides reliable and consistent results on real data.

2. Negative tests (permutation)

To ensure that the model is not learning by chance, a permutation test is performed:

The target values (PWV) are randomly shuffled, breaking their relationship with the features.

The model is re-evaluated on these permuted data, and the distribution of errors is analyzed.

III. Application and extension of the methodology to a new project: Chronic stress detection (experiment)

a. Project context

1. Preamble

At the end of the first month of the internship, which was devoted to studying the PWV-estimation pipeline, I was asked to take part in a related project led by a doctoral student in the team, focusing on the detection of chronic stress.

Stress, although useful in occasional situations (alertness, rapid reaction, adaptation), can become harmful when it persists over time. It is then associated with an increased risk of cardiovascular diseases (hypertension, heart attack), metabolic diseases (diabetes), and psychological disorders (depression, chronic anxiety). However, it is particularly difficult to distinguish chronic stress from other physiological states such as physical effort, a state of excitement or a simple increase in heart rate.

The objective of this project was to identify physiological markers specific to chronic stress by comparing it with other conditions that induce similar bodily reactions. To do this, an experimental protocol was designed to induce different types of stress in a controlled and reproducible manner:

- Cognitive stress: involving the participant's attention, memory and cognitive inhibition (e.g. memorizing harmless information followed by an unexpected instruction).
- Physiological stress: caused by painful or uncomfortable stimulation (e.g. immersing the hand in ice water).
- Physical stress: induced by increasing muscular effort on an exercise bike.

These three complementary categories make it possible to compare the impact of varied situations on the same physiological markers and to assess the specificity of the response associated with chronic stress.

2. Ethical and administrative aspects

Before each experimental session, a rigorous consent protocol was implemented in order to comply with research ethics and GDPR requirements:

- Signing an authorization form guaranteeing anonymity, the right to have data erased, and the participant's ability to withdraw at any time.
- A clear explanation to the participant of the nature of the physiological measurements and their purpose, using understandable, non-technical language.
- Secure storage of physiological data: in order to limit any risk of leakage, no identifying information or raw data could be sent by email or stored on personal computers or on the network. The files were stored exclusively on an encrypted, password-protected hard drive dedicated to the project.

These preliminary steps, although time-consuming, were essential in order to use the recordings afterwards within a scientific and legal framework while respecting participant confidentiality.

b. Equipment and physiological data collected

Each participant was equipped with a set of sensors designed to measure several physiological parameters simultaneously. This configuration was intended to collect multimodal data in order to better distinguish stress-related responses from other physiological states.

EEG - Electroencephalogram

- A cap with 32 electrodes positioned over different areas of the scalp measured electrical brain activity in real time.
- To improve conductivity, a cold conductive gel was applied, and the electrodes had small tips to anchor them slightly into the scalp.
- Light indicators (green, orange, red) showed the quality of the electrode-skin contact, which made adjustment easier before the experiment and reduced signal loss.



Figure 4: cap with the 32 EEG electrodes

ECG - Electrocardiogram

- Three additional electrodes (two on the pectoral muscles and one near the mediastinum) measured cardiac activity.
- In theory, the ECG was synchronized with the EEG via a trigger cable, which was supposed to send a time marker each time the device was switched on or an event occurred. This synchronization was intended to align the cardiac and brain signals precisely for subsequent analyses.
- In practice, however, the ECG proved particularly temperamental: electrodes came loose during movement, signal losses were frequent, and the sending of markers to the EEG did not always occur as expected. These malfunctions introduced gaps into the acquisition and required data-cleaning work.

Blood pressure measurement

- A finger blood-pressure monitor made it possible to continuously obtain peripheral blood pressure.
- Although accurate, this device was sensitive to movement, sometimes resulting in signal loss during the experiment.

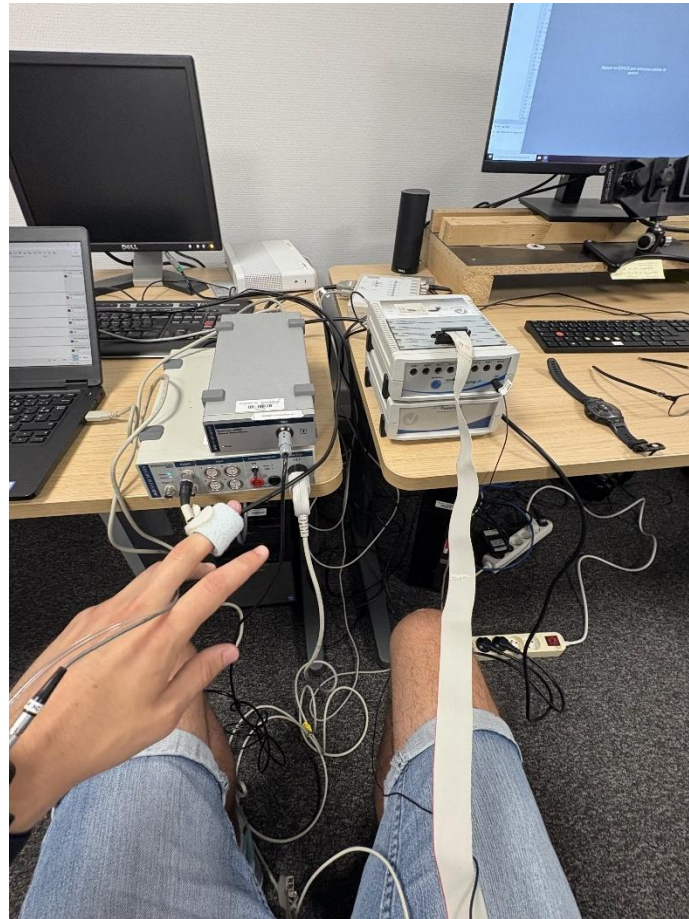


Figure 5: finger blood-pressure monitor

c. Experimental procedure

Each session first included a 120-second control phase at rest, used to calibrate the measurements and verify signal integrity. Three types of stress were then induced, separated by recovery phases:

1. Cognitive stress

Three harmless sentences were displayed successively for 60 seconds each (examples: "The first vacuum cleaner was invented in 1907", "Modern vacuum cleaners are quieter", "HEPA filters capture 99% of particles"). The participant thought they would have to memorize them for a future test, which directed the brain toward conventional attention and memorization mechanisms.

Suddenly, an unexpected instruction appeared: "Sing a song out loud for 60 seconds". By breaking expectations and adding a social and unexpected aspect, this instruction induced acute stress. The participant had to quickly find a song and sing it in public, with a running timer.

In my own case, I chose to sing "La Bohème" by Charles Aznavour, but it took me around thirty seconds to decide which song to perform, which illustrates the cognitive and emotional tension experienced by the volunteers.

This unusual situation caused, in most subjects:

- a marked increase in heart rate (measured by the ECG),

- significant variations in EEG activity, related to cognitive effort, social embarrassment and the unpredictability of the task.

A short 60-second recovery period followed this stage to allow physiological parameters to settle before moving on to the next phase.

2. Physiological stress

After a 60-second rest period, the participant had to immerse one hand in a container of ice water maintained at 4 °C for three minutes. This test, known as the Cold Pressor Test, is a standard protocol in experimental physiology for inducing marked activation of the sympathetic nervous system.

In practice, this test is particularly unpleasant: an intense burning sensation appears within the first few seconds, and keeping the hand in the ice water requires considerable mental effort. This constraint, combined with the pain and the initial surprise, makes the test a powerful trigger of physiological stress.

At the end of the three minutes, the participant removed their hand, dried it quickly and had a 60-second recovery period, which was essential to allow the physiological parameters to gradually return toward their baseline level before the next phase.

3. Physical stress

After an initial 60-second control phase intended to stabilize the measurements and check signal quality, the participant was asked to sit on an exercise bike. This simple change of position - leaving the chair to get onto the bike - frequently caused the ECG electrodes to come loose (movement of the cables, sweat or rubbing against clothing). This resulted in signal interruptions and lost segments in the cardiac recordings.

Once properly positioned, the participant had to follow a progressive exercise protocol:

- 7 successive stages, from 40 W to 160 W,
- increments of 20 W every two minutes,
- for a total duration of 14 minutes if the exercise was completed.

This protocol induced a progressive increase in heart rate and blood pressure, thereby reproducing a purely physical stress different from the previous phases (cognitive and physiological).

In practice, this was the most demanding test:

some participants, who were not sufficiently physically trained, could not reach the final stage,

sweating and movement increased the risk of electrodes coming loose, compromising the continuity of the ECG signal,

and general fatigue could lead to pauses or early termination.

These experimental constraints explain why the data collected during this phase are often noisier and more incomplete, but they nevertheless constitute a key element for comparing physiological responses to different types of stress.

IV. Application and extension of the methodology to a new project: Chronic stress detection (data processing)

To ensure the consistency and accuracy of the physiological measurements, the doctoral student supervised the development of a Python script by an intern in the team as part of her final-year project. The purpose of this script was to automatically control the entire experimental protocol, minimizing human errors and facilitating subsequent data processing.

The main functions of the script were as follows:

- Display of instructions and timers: each stage of the protocol (control, cognitive stress, physiological stress, physical stress) was shown on a screen, with precise counters to ensure that the planned durations were respected.
- Sending time markers to the EEG: at each phase transition, the script sent marker signals to precisely annotate the beginning and end of each stage. This allowed automatic segmentation of the EEG data. However, this was not the case for the ECG, which created a synchronization problem that I worked on.

a. Reading the data:

Each experimental session generates four main files:

EEG files:

- .vhdr: contains the general parameters relating to the EEG.
- .vmrk: history of the markers emitted during the experiment.
- .eeg: stores all EEG samples in encrypted form.

Only .vhdr and .vmrk can be read without specific software. The three files are complementary, so it is crucial not to separate them.

ECG file:

- .adicht: encodes all data relating to the ECG and blood pressure.

During the experiment, the signals could be viewed in real time using LabChart, but this licence-limited software did not allow advanced pre-processing. For the rest of the work, I used Python with two open-source libraries:

- **MNE-Python**: complete reading and analysis of EEG signals.
- **adstruments_sdk_python**: processing of ECG and blood-pressure signals.

```
import adi
import mne
```

For MNE, the data are read as follows:

```
raw_eeg = mne.io.read_raw_brainvision("manip_1sujet10_15_04.vhdr", preload=True) # load data

for ch_name in raw_eeg.info['ch_names']:
    eeg_data = raw_eeg.get_data(picks=ch_name[0])
```

Loading the files only requires specifying the vhdr file.

There are 32 EEG channels, each corresponding to a specific electrode placed on the scalp according to the international 10-20 system. These channels are:

['Fp1', 'F3', 'F7', 'FT9', 'FC5', 'FC1', 'C3', 'T7', 'TP9', 'CP5', 'CP1', 'Pz', 'P3', 'P7', 'O1', 'Oz', 'O2', 'P4', 'P8', 'TP10', 'CP6', 'CP2', 'Cz', 'C4', 'T8', 'FT10', 'FC6', 'FC2', 'F4', 'F8', 'Fp2']

Each channel makes it possible to observe the electrical activity of a specific brain region. For example:

- Fp1: located in the left frontal region near the forehead, it becomes active during tasks involving attention, working memory and certain aspects of cognitive reflection.
- Cz: central electrode at the top of the head, sensitive to motor and sensory activity.
- O1 / O2: left and right occipital regions, involved in vision and visual processing.
- F7 / F8: lateral frontal regions, associated with executive control, planning and sometimes emotions.
- T7 / T8: left and right temporal regions, important for hearing, speech processing and memory.

For ECG signals and the other physiological measurements obtained from the ADI device, operation is fairly similar to MNE for the EEG, but with a few particularities related to the architecture of .adicht files.

Each file contains several channels corresponding to the measurements recorded during the experiment. In our case, there are 5 main channels:

- ECG: cardiac activity
- BP: blood pressure (not separated into systolic/diastolic)
- SBP: systolic blood pressure
- DBP: diastolic blood pressure
- MBP: mean blood pressure

For each channel, several independently recorded segments may exist (indexed from 1). This is due to automatic interruptions of the device, often caused by:

- a connection that is too fragile on the patient
- excessive movement by the volunteer
- or a temporary malfunction of the machine

These interruptions therefore generate segments of variable duration, which must be reconstructed in order to obtain a continuous signal. Naive concatenation of all segments is performed as follows:

```
adi_file = adi.read_file('manip1_sujet16_23_04.adicht') # load data

for channel in adi_file.channels:
    if (channel.n_samples[0] > 0): # not all channels are used
        ecg_data = channel_data = np.concatenate([channel.get_data(k) for k in range(1, len(channel.n_samples)+1)])
```

b - Marker management and data segmentation

For automatic signal processing to be possible, each data segment must be labelled according to the phase of the experiment to which it corresponds. This makes it possible to link each portion of the EEG or ECG signal to the precise experimental stage (control, cognitive, physiological or physical stress).

Fortunately, the PsychoPy script used to display instructions to the participant was designed to send markers to the EEG at each phase transition. Communication takes place through a serial port:

```
import serial
port = serial.Serial('COM4', baudrate=9600, timeout=1)
```

COM4 corresponds to the port connected directly to the EEG.

The script sends 22 indexed markers in byte form:

```
port.write(bytes([1])) -> port.write(bytes([22]))
```

These markers are then listed in the .vmrk file associated with the EEG. Example:

Mk1=New Segment,1,1,0,20250423160401733694	Mk11=Stimulus,S 10,759044,1,0
Mk2=Stimulus,S 1,260833,1,0	Mk12=Stimulus,S 11,942168,1,0
Mk3=Stimulus,S 2,382888,1,0	Mk13=Stimulus,S 12,1064246,1,0
Mk4=Stimulus,S 3,385207,1,0	Mk14=Stimulus,S 14,1118107,1,0
Mk5=Stimulus,S 4,446235,1,0	Mk15=Stimulus,S 15,1240164,1,0
Mk6=Stimulus,S 5,507268,1,0	Mk16=Response,R 1,1362241,1,0
Mk7=Stimulus,S 6,568299,1,0	Mk17=Stimulus,S 1,1484319,1,0
Mk8=Stimulus,S 7,629328,1,0	Mk18=Stimulus,S 2,1606430,1,0
Mk9=Stimulus,S 8,690357,1,0	Mk19=Stimulus,S 3,1728558,1,0
Mk10=Stimulus,S 9,698010,1,0	Mk20=Stimulus,S 4,1850685,1,0
	Mk21=Stimulus,S 5,1972763,1,0

Because of the lack of documentation and the limited experience of some collaborators, I had to reverse-engineer the 30 .vmrk files generated during the experiment. The objective was to understand the exact role of each marker, a crucial step for EEG data pre-processing.

The analysis made it possible to identify the marker format fairly quickly, which is as follows:

First, the format of these markers can be identified quite easily:

[Type, no., position among the samples]

- Type: indicates whether the marker corresponds to a stimulus, a response or a new segment.
- Number: identifies the marker in the sequence of the experiment.
- Position: the index of the EEG sample corresponding to this marker.

With a sampling frequency of 1000 Hz, these positions can be converted into time, making it possible to estimate the delays between each event precisely.

By combining this information with the PsychoPy instruction code, which specifies the durations and intervals between stimuli, we were able to establish a reliable correspondence between the markers emitted by the script and those recorded by the EEG, which is essential for correctly segmenting the experimental data.

Timeline of the experiment

The complete sequence of experimental stages, as recorded by the EEG markers, can be summarized as follows:

1 -> 2 Control period 120s	10 -> 11 180s immersion
2 -> 3 Start of SSST random pressure delay	11 -> 12 120s recovery time
3 -> 4 60s question display time	12 -> 13 random Cyclo delay
4 -> 5 60s question display time	13 -> 14 instantaneous! hence disappearance of 13!!!!
5 -> 6 60s question display time	14 -> 15 120s cycling stage
6 -> 7 60s display time for the stressful question	15 -> 0(1) 120s cycling stage
7 -> 8 60s recovery period	0(1) -> 1 120s cycling stage
8 -> 9 random pressure delay for CPT	2 -> 3 120s cycling stage
9 -> 10 60s control period	3 -> 4 120s cycling stage
	4 -> 5 120s cycling stage
	5 -> 6 120s final recovery phase

Analysis of marker indexing

During the study of the VMRK files, several anomalies in marker indexing were identified:

1. 4-bit encoding

- The markers appear to be encoded on 4 bits, unlike the 8 bits sent by the Python script.
- This limitation is not critical: when the counter exceeds 15 (the maximum value on 4 bits), the system automatically resets the counter by truncating the most significant bits.

2. Aberration on the 16th marker

- Bit 0 of marker 16 is interpreted as a machine error of type Response with index 1.
- However, it is actually a valid marker emitted by the instruction script. This anomaly must be taken into account during pre-processing so that errors are not confused with genuine markers.

3. Absence of marker 13

- In some files, marker 13 does not appear.
- This comes from an error in the instruction code: two bytes (13 and 14) are written consecutively with no delay.
- Because the read buffer is too small, byte 13 is overwritten by byte 14 before it can be read, explaining why it disappears.

These observations are important because they affect the correspondence between physiological events and markers, which is essential for the segmentation and analysis of EEG and ECG signals.

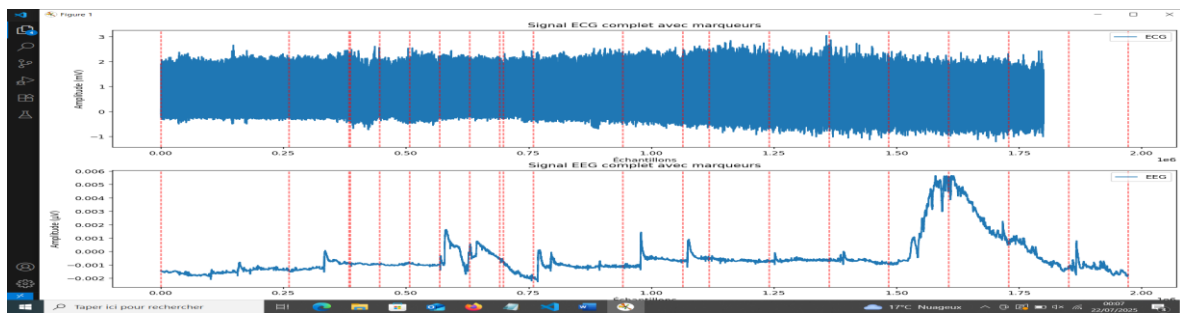


Figure 5: segmentation of instantaneous BPM from the EEG markers without taking synchronization into account

c - Synchronization of EEG and ECG signals

The experiment uses multimodal data measured with several independent instruments:

- EEG for brain activity,
- ECG for cardiac activity and blood pressure.

However, only the EEG signals are automatically marked by the events of the experiment (VMRK markers). To make full use of the data, it is necessary to synchronize the ECG signals with the EEG in order to correctly segment and label each phase of the protocol.

Synchronization principle

- When the experiment was designed, the ECG was intended to be connected to the EEG so that each time the ECG was switched on, a marker would be generated in the EEG.
- This marker indicates the start of ECG acquisition, making it possible to bring the ECG and EEG data into temporal alignment.
- This method makes it possible to correct ECG dropouts or interruptions by precisely aligning the valid segments with the corresponding EEG signal.

Mk1=New Segment,,1,1,0,20250506111359207325	Mk12=Stimulus,S 10,548357,1,0
Mk2=Response,R 8,11966,1,0	Mk13=Stimulus,S 11,731462,1,0
Mk3=Stimulus,S 1,44526,1,0	Mk14=Stimulus,S 12,853558,1,0
Mk4=Stimulus,S 2,166600,1,0	Mk15=Stimulus,S 14,951246,1,0
Mk5=Stimulus,S 3,170971,1,0	Mk16=Stimulus,S 15,1073417,1,0
Mk6=Stimulus,S 4,232016,1,0	Mk17=Response,R 1,1195543,1,0
Mk7=Stimulus,S 5,293062,1,0	Mk18=Stimulus,S 1,1317605,1,0
Mk8=Stimulus,S 6,354091,1,0	Mk19=Stimulus,S 2,1439699,1,0
Mk9=Stimulus,S 7,415139,1,0	Mk20=Stimulus,S 3,1561825,1,0
Mk10=Stimulus,S 8,476170,1,0	Mk21=Stimulus,S 4,1684020,1,0
Mk11=Stimulus,S 9,487334,1,0	Mk22=Stimulus,S 5,1806131,1,0

In the majority of recordings, the marker sent when the ECG is switched on appears as a Response with index 8 in the VMRK file.

- We consider as simple cases all files containing only Response 8 markers for ECG switch-ons and in which the other markers are correctly ordered.
- Even if several Response 8 markers are present, indicating that the ECG experienced dropouts or temporary interruptions during the experiment, it remains possible to synchronize the ECG and EEG signals correctly.

To illustrate the issue of interruptions in the ECG, let us consider a concrete recording from a simple case.

```
adi_file = adi.read_file('manip1_sujet8.adicht')
print(f'{adi_file.channels[2].name} : {adi_file.channels[2].n_samples}')
```

Output: ECG: [1522938, 678888]

The ECG track consists of two continuous segments.

The first segment contains 1,522,938 samples.

The second segment contains 678,888 samples.

Between these two segments, there is an interruption whose exact duration is unknown, caused by an electrode coming loose or by participant movement.

Here is the associated vmrk file:

Mk1=New Segment,1,1,0,20250411100818394628	Mk14=Stimulus,S 10,749478,1,0
Mk2=Response,R 8,51220,1,0	Mk15=Stimulus,S 11,932584,1,0
Mk3=Response,R 8,92329,1,0	Mk16=Stimulus,S 12,1054647,1,0
Mk4=Response,R 8,137452,1,0	Mk17=Stimulus,S 14,1498892,1,0
Mk5=Stimulus,S 1,181937,1,0	Mk18=Stimulus,S 15,1621005,1,0
Mk6=Stimulus,S 2,304007,1,0	Mk19=Response,R 8,1678803,1,0
Mk7=Stimulus,S 3,314246,1,0	Mk20=Response,R 1,1743130,1,0
Mk8=Stimulus,S 4,375274,1,0	Mk21=Stimulus,S 1,1865243,1,0
Mk9=Stimulus,S 5,436306,1,0	Mk22=Stimulus,S 2,1987385,1,0
Mk10=Stimulus,S 6,497336,1,0	Mk23=Stimulus,S 3,2109497,1,0
Mk11=Stimulus,S 7,558381,1,0	Mk24=Stimulus,S 4,2231590,1,0
Mk12=Stimulus,S 8,619413,1,0	Mk25=Stimulus,S 5,2353668,1,0
Mk13=Stimulus,S 9,688447,1,0	

By using the EEG markers to estimate the total duration between two successive ECG switch-ons, synchronization can be performed even when an interruption occurs. For example, considering the last R8 marker before the beginning of the experimental phase (MK4) and the following R8 marker (MK19), we obtain:

$$1678803 - 137452 = 1\,541\,351 \text{ samples}$$

However, the first ECG segment contains only 1,522,938 samples. This indicates an interruption of 18,413 samples. To compensate for this interruption, the missing segment can simply be filled with zeros or None values, making it possible to reconstruct an ECG signal aligned with the EEG. This process can be repeated for all successive interruptions, ensuring reliable synchronization despite the dropouts.

Applying this principle to the data from the previous participant gives the following result:

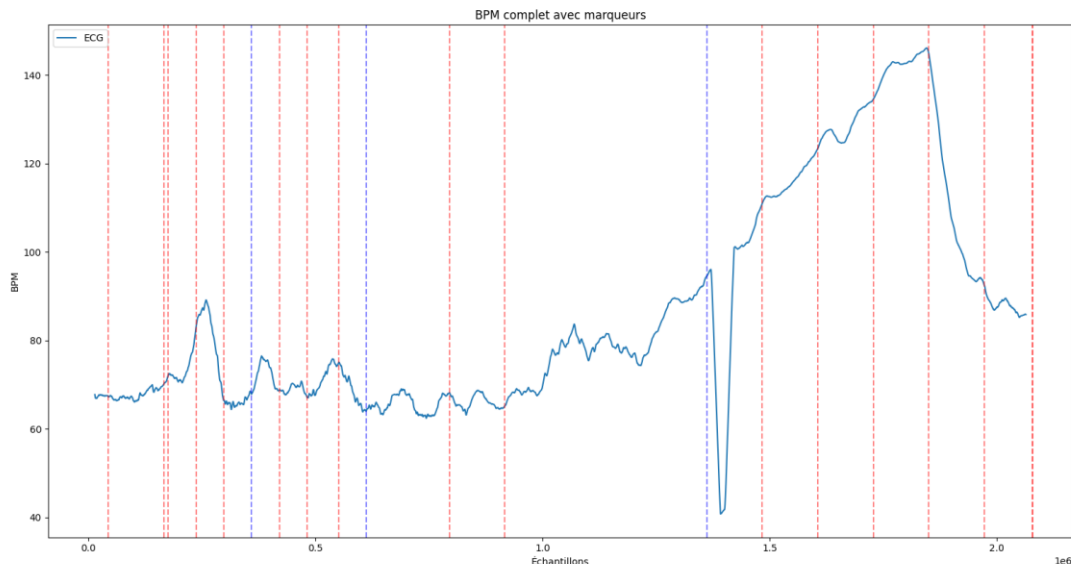


Figure 6: segmentation of instantaneous BPM from the EEG markers using the synchronization principles stated above

In blue, the three stress phases; in orange, the others. We filled the gaps with zeros and checked synchronization using instantaneous BPM. And then... major disappointment: the stress phases do not match the peaks at all! Is this an error in the method, or are the data simply impossible to align correctly? Frankly, with data collection this messy, it is almost mission impossible. Even after a month of tearing my hair out over these data, it is still the same puzzle.

d - Improper cases

Unfortunately, not all collected VMRK files correspond to simple cases. In particular, two other situations can be distinguished.

The first type is what we call pathological cases. These files remain close to simple cases, but have one particular feature: the index of the ECG switch-on marker is not constant. Whereas it is normally fixed at 8 in simple cases, it can take apparently random values while retaining the Response type.

The main difficulty with this type of file lies in distinguishing between a genuine ECG switch-on and the 16th marker, which is itself encoded as a Response with index 1. Without careful analysis, these events can easily be confused, which complicates the synchronization of multimodal signals.

Example of a vmrk file from a pathological case:

Mk1=New Segment,1,1,0,20250410120727616175	Mk19=Stimulus,S 10,754063,1,0
Mk2=Response,R 9,17786,1,0	Mk20=Stimulus,S 11,937152,1,0
Mk3=Response,R 1,17886,1,0	Mk21=Stimulus,S 12,1059213,1,0
Mk4=Response,R 9,53401,1,0	Mk22=Response,R 8,1304129,1,0
Mk5=Response,R 1,53501,1,0	Mk23=Response,R 8,1317249,1,0
Mk6=Response,R 9,142501,1,0	Mk24=Response,R 8,1335364,1,0
Mk7=Response,R 1,142601,1,0	Mk25=Stimulus,S 14,1488045,1,0
Mk8=Response,R 9,200431,1,0	Mk26=Response,R 8,1506744,1,0
Mk9=Response,R 1,200531,1,0	Mk27=Stimulus,S 15,1610103,1,0
Mk10=Stimulus,S 1,243505,1,0	Mk28=Response,R 1,1732200,1,0
Mk11=Stimulus,S 2,365568,1,0	Mk29=Stimulus,S 1,1854259,1,0
Mk12=Stimulus,S 3,371439,1,0	Mk30=Stimulus,S 2,1976336,1,0
Mk13=Stimulus,S 4,432468,1,0	Mk31=Stimulus,S 3,2098414,1,0
Mk14=Stimulus,S 5,493498,1,0	Mk32=Stimulus,S 4,2220475,1,0
Mk15=Stimulus,S 6,554530,1,0	Mk33=Response,R 9,2330988,1,0
Mk16=Stimulus,S 7,615576,1,0	Mk34=Response,R 1,2331088,1,0
Mk17=Stimulus,S 8,676607,1,0	Mk35=Stimulus,S 5,2342536,1,0
Mk18=Stimulus,S 9,693034,1,0	Mk36=Stimulus,S 6,2464596,1,0

The final type of file, and by far the most complex to process, corresponds to cases without a synchronization marker. In these files, the overall chronology of the experiment is respected, and all stages appear correctly in the EEG and ECG data. However, a Response-type marker is completely absent, which complicates segmentation and signal labelling.

Without this marker, it becomes difficult to identify precisely the beginning or end of an experimental phase in the ECG, making synchronization with the EEG more delicate. Although the temporal structure of the other markers makes it possible to estimate events approximately, the absence of the reference marker prevents complete automation of pre-processing.

For these files, two strategies were considered during meetings with the doctoral student:

1. Signal-independent method: assume that the ECG and EEG were switched off a few seconds apart and align the final values of the two signals to synchronize the instruments approximately.
2. Method based on temporal interpolation: reconstruct the missing marker based on the expected duration between adjacent events, then adjust the ECG segments accordingly.

These cases remain exceptional and require manual or semi-automated processing, making them a major challenge for systematic data analysis.

Example of a vmrk file without a marker

Mk1=New Segment,1,1,0,20250319193828554439	Mk13=Stimulus,S 12,1064600,1,0
Mk2=Stimulus,S 1,259600,1,0	Mk14=Stimulus,S 14,1253104,1,0
Mk3=Stimulus,S 2,381660,1,0	Mk15=Stimulus,S 15,1375206,1,0
Mk4=Stimulus,S 3,384182,1,0	Mk16=Response,R 1,1497316,1,0
Mk5=Stimulus,S 4,445226,1,0	Mk17=Stimulus,S 1,1619378,1,0
Mk6=Stimulus,S 5,506256,1,0	Mk18=Stimulus,S 2,1741453,1,0
Mk7=Stimulus,S 6,567302,1,0	Mk19=Stimulus,S 3,1863531,1,0
Mk8=Stimulus,S 7,628333,1,0	Mk20=Stimulus,S 4,1985675,1,0
Mk9=Stimulus,S 8,689365,1,0	Mk21=Stimulus,S 5,2107787,1,0
Mk10=Stimulus,S 9,698393,1,0	Mk22=Stimulus,S 6,2229848,1,0
Mk11=Stimulus,S 10,759432,1,0	
Mk12=Stimulus,S 11,942524,1,0	

For time-constraint reasons, we will not process cases without synchronization markers that also contain interruptions in the ECG. During a meeting with the doctoral student in charge of the project, one method was discussed for handling these particular cases: assuming that the ECG and EEG stopped a few seconds apart, then matching the last recorded ECG values with those of the EEG. This approach makes it possible to approximate signal synchronization, but introduces uncertainty related to the actual delay between the shutdown of the two devices, making it insufficiently reliable for using these data in a machine-learning algorithm.

Another method considered would consist of identifying several characteristic segments in the ECG, for example the three stress periods, which are fairly visible when observing instantaneous BPM changes, and then realigning them relative to the EEG data. However, this approach remains delicate because the physiological response time to stress varies greatly from one patient to another, making precise and reproducible alignment difficult.

Conclusion

In conclusion, this part of the data processing was particularly frustrating. The lack of rigor in data collection, combined with the absence of effective supervision by the doctoral student, who left the intern to carry out the experiments alone, resulted in a majority of files being partially or completely unusable. ECG-EEG synchronization, although crucial for multimodality, was often compromised. Although it remains possible to use the EEG data alone, this approach has already been widely studied by other teams, so the scientific value of multimodality is greatly reduced.

On a personal level, spending an entire month struggling with these files left a bitter taste: a great deal of time and energy were spent for limited benefit. The initial objective of doing AI work could ultimately not be achieved as I had hoped.

However, this experience also allowed me to discover the world of research, with its constraints and requirements, and to become familiar with very interesting theoretical methods. I particularly enjoyed studying visibility graphs, as well as the EBM approach, and I would have liked to apply these tools to the features extracted from the signals. Unfortunately, as I was in Montreal for a double-degree programme, time did not allow me to continue this part, which remains a source of regret but also an objective for future projects.

Thus, despite the difficulties encountered, this phase was a valuable learning experience both scientifically and personally, giving me a concrete insight into experimental research and its methodological requirements. It will serve as a reference for my future projects and my choices of working methods.